

Hanyang Wang

Data Efficiency · LLM Post-Training · Agentic RL-OPD · Agent Harness · LLM Data Synthesis

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EDUCATION

The University of Chicago | M.S. in Data Science (Artificial Intelligence)

2025.09 - 2026.12

1. **Research Interests:** Data Efficiency, LLM Post-Training, Agentic RL, Agent Harness, and LLM Data Synthesis.

The University of Hong Kong | Exchange Student in Statistics

2023.09 - 2024.01

1. **Selected Coursework:** Statistical Risk Management, Statistical Machine Learning, Statistical Inference, Stochastic Processes, and Linear Statistical Analysis.

University of Nottingham | B.Sc. in Statistics

2020.09 - 2024.07

1. **Selected Coursework:** Numerical Computation, Probability Models and Methods, Statistical Modeling, Mathematical Analysis, Time Series Analysis, and Applied Statistical Modeling.
2. **Honors & Awards:** Turing Scheme Mobility Grant (UK government funded, £2,560); University of Nottingham Study Abroad Scholarship (£4,600); KAUST Visiting Student Research Program Fellowship (\$1,000/month, 2023).

EXPERIENCE

Meituan | LongCat · [Beidou] Foundation-Model Agentic Coding

2026.06 - Present

1. **DivOPD / Agentic OPD:** Led DivOPD, a learner-side batch composer that exposed update-level credit-allocation failure in vanilla OPD: **a 64-turn batch covered only 4-8 effective rollouts, one rollout contributed 24%-40% of tokens, and up to 27.8% of turns lacked valid teacher supervision**. Designed validate-cover-focus: filter zero-gradient turns with a validity gate, ensure rollout coverage via trajectory-first sampling with a per-rollout cap, then prioritize high-learning-need turns using sampled forward KL (raw_kl_sum). With the loss, model, optimizer, and rollout procedure unchanged, raised mean peak success rate from **77.4% to 83.7%** across six 1.5B-7B settings on three environments and achieved **1.78× average time-to-target speedup (2.22× max)**.
2. **HBR-GSPO / Agentic RL:** Designed a two-level denoised advantage estimator for Coordinator-SubAgent hierarchical search. Replayed candidate actions from the same state, marginalized sub-agent retrieval noise through within-action Monte Carlo resampling, and performed group-relative comparison across actions to separate search-direction value from channel randomness in outcome rewards. Built paired finish/continue controls to learn a stopping boundary, suppressing over-search without a hand-tuned length penalty.
3. **LongCat2.5 Data Synthesis:** Built executable agentic task environments on Harbor, modeled on Terminal-Bench, for SWE-SQL and OfficeQA-Pro. Developed an auditable quality pipeline spanning multi-source document sampling, K3 question generation and solving, independent validation, and programmatic answer recomputation, with structured-rubric filtering. Created **9K agentic tasks** and distilled K3 trajectories to fine-tune **Flash-26B-DSA**, improving OfficeQA-Pro Full score from **46.6% to 60.2% (+13.5 pp)**.

Tencent | CDG Advertising (WeChat Official Accounts) · RL for Recommendation

2025.06 - 2025.09

1. **DRL-PUT Framework:** Formulated multi-objective utility-weight tuning for WeChat Official Accounts/Channels ad ranking as a deep-RL decision problem, addressing manual tuning cost, user heterogeneity, and seasonal drift. Benchmarked continuous-control methods including DDPG and SAC before selecting REINFORCE, avoiding brittle value-function estimation under sparse, high-variance rewards.
2. **Personalized Multi-Objective Reward:** Built a reward combining platform revenue (Estimated Revenue) with user value (weighted CTR/CTR30/CVR), with campaign-specific weighting for click-through, conversion, and impression objectives. Bucketed predicted hyperparameters by historical CTR/CVR/CPM and verified that the policy learned personalized decisions, increasing w_click and reducing w_conversion for click-prone users.
3. **Production Deployment & Online Validation:** Combined continuous-action discretization with semantically related weight grouping to compress the action space to $\sim 10^3$ choices and meet real-time inference SLAs at 100M+ MAU. Shipped the full offline-training-to-online-serving stack; a 25% traffic A/B test at billion-impression scale delivered **CTR +9.7%, CTR30 +7.7%, and CVR +1.26%** with no significant regression in core revenue metrics.

Microsoft | Microsoft AI · Foundation-Model Post-Training

2025.01 - 2025.06

1. **Text2Grad Framework:** Proposed and implemented Text2Grad, converting natural-language feedback into span-level differentiable gradient signals to address reward sparsity and coarse credit assignment in PPO/RLHF. Built a semantic-alignment $\rightarrow$ reward-mapping $\rightarrow$ gradient-propagation loop that localizes critiques to erroneous token/reasoning spans. Outperformed PPO, DPO, and PRM-PPO across summarization, code generation, and open-domain QA: **BLEU +25.3%, HumanEval+ +3.6, AlpacaEval +2.3, and 22% faster convergence**.
2. **Multi-Task RLHF Data:** Designed a GPT-4o CoT-based Dual-Feedback annotation pipeline (Critique $\rightarrow$ Span-Level Alignment) across SLF5K, UltraFeedback, and KodCode. Produced high-fidelity span-level labels with $<2.5\%$ unmatched rate and 82%-94% human alignment, while reducing cost by 85%-90% versus dense token-level annotation; supplied cross-task supervision for NL-Gradient training.
3. **MLLM Vision-Language Alignment:** Fine-tuned Qwen2.5-VL-3B/7B with LoRA and full-parameter training for region-level vision-language alignment in GUI understanding, extending fine-grained feedback from text-only to multimodal settings.

1. **Automated Evaluation Benchmark:** Led a hybrid automated benchmark for Code-IDE, combining Claude-3.5 chain-of-thought reasoning and self-reflection to score code readability, style conformance, and intent alignment. Evaluated 8-10 candidate models in parallel, achieved **0.85+ Spearman correlation** with human rankings, and reduced evaluation turnaround from 2-3 days to hours.
2. **Structured Reverse-Distillation Data:** Designed a Claude-3.5 multi-round CoT self-correction pipeline for frequent production bad cases, automatically attributing syntax, logic, API-misuse, and edge-case failures. Produced traceable error-type -> repair-path -> reasoning-alignment labels for full-parameter SFT and DPO preference alignment.
3. **SFT & DPO Data Strategy:** Built a production-log training set centered on long context, cross-function calls, and boundary conditions. Kept only 1-2 boundary failures in each DPO negative and enforced a positive-negative test_case_accuracy gap $\geq 30\%$, preserving a strong learning signal while suppressing reward hacking.
4. **Step-Level DPO:** Applied preference alignment at each reasoning step, rather than only the final output, for structured coding tasks. Used staged curriculum training to control growth of the easy-sample ratio, mitigating late-stage DPO overconfidence collapse and improving robustness on difficult boundary cases.

Xiaohongshu | Hi-Lab · Foundation-Model Evaluation

2024.07 - 2024.09

1. **Complex-Task Evaluation:** Contributed to the OpenCompass open-source community by integrating CMath and CSEM reasoning/semantic-understanding datasets and refactoring the preprocessing pipeline and query engine for **3 $\times$ evaluation throughput**. Contributed to a general foundation-model benchmark spanning language understanding, knowledge QA, complex reasoning, code generation, and instruction following.
2. **Multimodal Evaluation Benchmark:** Designed an MLLM benchmark spanning image-text similarity, image-aesthetic scoring, and cross-modal structured extraction (OCR + semantics).

RESEARCH EXPERIENCE

BiPACE for Long-Horizon LLM Agents

2026.04 - 2026.06

1. Identified a state-action credit mismatch in group-based RL for long-horizon LLM agents, caused by overly fine observation-hash grouping and coarse action credit assignment. Proposed BiPACE, requiring neither a critic nor additional rollouts, and achieved SoTA on ALFWorld, WebShop, and TextCraft.
2. **BiGPO:** Clustered actor hidden-state representations by cosine similarity to form approximate-bisimulation state groups. Replaced exact observation hashing with behavioral-equivalence grouping so equivalent states could share learning signals.
3. **PACE:** Constructed an action-conditioned local peer baseline within each state group to estimate counterfactual advantage and provide finer action-level credit assignment. On ALFWorld with Qwen2.5-7B, improved success rate from GiGPO's 90.8% to **97.1 $\pm$ 0.9%**, with higher sample efficiency and final performance.

SkillRL

2025.11 - 2026.02

1. Validated SkillRL on ALFWorld, WebShop, and Search-Augmented QA, outperforming strong baselines by $\sim 15.3\%$ overall. On ALFWorld, increased success rate from GRPO's 77.6% to **89.9%**, demonstrating stable gains and cross-task generalization from skill abstraction in sparse-reward agent RL.
2. **SkillRL:** Addressed noise and context redundancy in raw trajectory memory through experience-based distillation, abstracting interaction traces into a hierarchical SkillBank that separates general from task-specific skills. Structured skills replaced redundant trajectory memory, improving context efficiency and pass@1 without introducing complex credit assignment.
3. **Recursive Skill Evolution:** Continuously detected failure modes missing from the SkillBank via validation failures, then generated or revised skills automatically. Policy failures triggered skill updates, and new skills fed back into policy exploration, creating a co-evolution loop between the SkillBank and agent policy.

Text2Grad

2025.01 - 2025.06

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PUBLICATIONS & PREPRINTS

* Equal contribution; † Corresponding author. All other papers are independent first-author works. Full list and links: [Google Scholar](#).

- [1] **Hanyang Wang**, Zeyuan Liu, Jingqing Ruan, Zhengyu Chen, Zhizhao Zeng. “**DivOPD: Rollout-Aware Batch Composition for On-Policy Distillation of Multi-Turn Agents**” · ICLR 2027 Under Review · [arXiv](#)
- [2] **Hanyang Wang**, Weijieying Ren, Yuxiang Zhang, Ding Cao, Zhizhao Zeng, Ke Zeng, Tianxiang Zhao. “**BiPACE: Bisimulation-Guided Policy Optimization with Action Counterfactual Estimation for LLM Agents**” · AAAI 2026 Under Review · [arXiv](#) · [code](#)

- [3] **Hanyang Wang**, Mingxuan Zhu. “**The Detection-Extraction Gap: Models Know the Answer Before They Can Say It**” · NeurIPS 2026 Under Review · [arXiv](#) · [code](#)
- [4] Peng Xia*, Jianwen Chen*, **Hanyang Wang***, Jiaqi Liu, Kaide Zeng, Yu Wang, Siwei Han, Yiyang Zhou, Xujiang Zhao, Haifeng Chen, Zeyu Zheng, Cihang Xie, Huaxiu Yao. “**SkillRL: Evolving Agents via Recursive Skill-Augmented Reinforcement Learning**” · ICLR 2026 LLA · [arXiv](#) · [code](#)
- [5] **Hanyang Wang**, Lu Wang†, Chaoyun Zhang, Tianjun Mao, Si Qin, Qingwei Lin, Saravan Rajmohan, Dongmei Zhang. “**Text2Grad: Reinforcement Learning from Natural Language Feedback**” · ICLR 2026 Main Conference · [arXiv](#) · [code](#)
- [6] **Hanyang Wang**, Hao Zhou, Sibao Cheng. “**Dynamical System Prediction from Sparse Observations Using Deep Neural Networks with Voronoi Tessellation and Physics Constraint**” · Computer Methods in Applied Mechanics and Engineering 432 (2024), 117339 · [DOI](#)
- [7] **Hanyang Wang**. “**Prob-cGAN: A Probabilistic Conditional Generative Adversarial Network for LSD1 Inhibitor Activity Prediction**” · Preprint · [arXiv](#)

Ongoing Work:Core experiments are complete for two papers on agentic OPD and search-agent RL; manuscripts are in preparation.